Supplemental Material

Part 1: Additional Sample and Treatment Information

Bern Randomized Trial

In the Bern parent trial, patients were randomized to Psychological Therapy (PT) with emotion-focused add-ons (PT + EFT; *n* = 52) or PT with self-regulation-focused add-ons (PT + SR; *n* = 52). The analytic sample comprised 101 patients, including 50 assigned to PT + EFT and 51 assigned to PT + SR.

Patients received outpatient psychotherapy based on PT, an integrative cognitive-behavioral treatment approach developed in the Bern tradition and grounded in Grawe's General Psychotherapy framework (Grawe, 2007; Grawe et al., 1990). PT combines cognitive-behavioral interventions with empirically supported techniques from other therapeutic traditions within a unified framework of psychological functioning and mechanisms of change. Treatment is guided by individualized case formulations based on Plan Analysis (Babl et al., 2016; Caspar, 2022; Caspar et al., 2023). Both conditions shared the overarching PT framework but differed in the systematic integration of either emotion-focused or self-regulation-focused components.

Treatment in the parent trial was delivered at a Swiss university outpatient clinic by 38 psychologists who had completed at least 1.5 years of postgraduate psychotherapy training. Therapists received condition-specific training and supervision with equivalent amounts of training and supervision across conditions. Further details regarding the treatment rationale, intervention procedures, therapist training, and treatment integrity are reported elsewhere (Babl et al., 2016; Caspar et al., 2023).

Zurich Randomized Trial

In the Zurich parent trial, major depressive disorder was confirmed at a face-to-face diagnostic interview. After providing informed consent, patients were randomly assigned by an external researcher to Exposure-Based Cognitive Therapy Revised (EBCT-R; *n* = 77) or

Cognitive-Behavioral Therapy (CBT; *n* = 72). The analytic sample comprised 145 patients, including 74 assigned to EBCT-R and 71 assigned to CBT.

Both conditions comprised up to 22 weekly sessions structured into three treatment phases. Phase 1 in both conditions covered psychoeducation, an individualized treatment rationale, and behavioral activation. EBCT-R further included work on interrupting rumination and avoidance patterns, building coping and problem-solving skills, and mindfulness practices to support distress tolerance and more adaptive engagement with emotions.

The conditions differed most clearly during Phase 2: CBT, based on the German manual by Hautzinger (2003), targeted dysfunctional cognitions and assumptions through cognitive restructuring and reality testing, whereas EBCT-R activated patients' depressive networks by recalling specific experiences linked to views of the self as failing, defective, or worthless, followed by processing of the activated material through emotion-focused techniques such as two-chair and empty-chair dialogues within an exposure framework.

Phase 3 in both conditions addressed relapse prevention, whereas EBCT-R additionally emphasized positive growth through goal setting, clarification of meaning and purpose, and development of a healthier view of the self.

Treatment in the parent trial was delivered by 25 master's-level psychologists in postgraduate psychotherapy training. Therapists received extensive training in both treatment approaches before the trial. Further details regarding treatment rationale, procedures, therapist training, and treatment integrity are reported in Grosse Holtforth et al. (2019).

Trier Routine-Care Sample

The Trier sample was drawn from patients treated at the outpatient clinic at Trier University between 2017 and 2026. Before the pretreatment assessment, patients receive detailed information about the research program, including its aims and assessment schedule, and provide written informed consent.

Diagnostic interviews were conducted by trained independent clinicians, videotaped, and reviewed in expert consensus meetings involving four senior clinicians.

Treatment consisted of personalized integrative CBT and was delivered by both licensed clinicians and therapists in postgraduate training programs. Therapists in training had at least 1.5 years of clinical experience and received regular individual or group supervision from senior clinicians. Treatment was typically delivered in weekly sessions and allowed flexibility in treatment strategies and techniques.

Pretreatment, posttreatment, and regular interval assessments every five sessions were administered through a web-based feedback system. Computerized feedback on patient status and treatment progress was provided through the Trier Treatment Navigator, a web-based system integrating outcome monitoring and clinical decision-support tools (Lutz et al., 2019).

Part 2: Additional Measures Information

Motivational Incongruence

Bern and Zurich trials: S-INC.

The original version of the Incongruence Questionnaire (INC) comprises 23 subscales, including 14 assessing approach incongruence and nine assessing avoidance incongruence, which yield summary scores for approach, avoidance, and total incongruence. The S-INC consists of 23 items, each selected as the most discriminating item from one of the 23 original subscales (Grosse Holtforth & Grawe, 2003).

Trier sample: INC-10

The INC-10 is a 10-item short form of the original INC, comprising six items assessing approach incongruence and four assessing avoidance incongruence. Items were selected based on their association with the original INC total score and measures of psychological symptoms and interpersonal functioning, as well as their retest reliability and sensitivity to change (Lutz et al., 2006).

General Psychological Distress

Because the instruments use different labels for the total score, we refer to the resulting scores as indices of general psychological distress, with higher scores indicating greater distress.

Bern trial: BSI

The Brief Symptom Inventory (BSI; Derogatis & Melisaratos, 1983; Franke, 2000) is a 53-item short form of the Symptom Checklist-90-R (SCL-90-R; Derogatis, 1977; Franke, 2002). The BSI retains the nine primary symptom dimensions of the original instrument: somatization, obsessive-compulsive symptoms, interpersonal sensitivity, depression, anxiety, hostility, phobic anxiety, paranoid ideation, and psychoticism. Four additional items contribute to the global indices.

Zurich trial: SCL-K-9

The Symptom-Checklist-K-9 (SCL-K-9; Klaghofer & Brähler, 2001) contains one item from each of the nine primary symptom dimensions of the SCL-90-R. Within each dimension, the item showing the strongest association with the Global Severity Index (GSI) of the full SCL-90-R was selected.

Trier sample: HSCL-11

The 11-item Hopkins Symptom Checklist (HSCL-11; Lutz et al., 2006) was developed through a step-wise item-selection procedure. Items were preselected based on their depression- and anxiety-related content and were then selected based on their associations with relevant SCL-90-R subscales and a measure of depressive symptoms, as well as their retest reliability and sensitivity to change.

At pretreatment, the BSI was administered instead of the HSCL-11. The corresponding BSI items in the source dataset had been linearly transformed as part of the standard procedure in the Trier setting to increase comparability with the HSCL-11 response

format.[1] The HSCL-11 total score was computed as the mean of all 11 items at each assessment.

Part 3: Tests of Temporal Mean Trends

To examine whether temporal mean trends were present, we first inspected the wave means visually. For each sample, we plotted the means and 95% confidence intervals of motivational incongruence and general psychological distress at each wave (Figure S1).

We then tested the trends formally using simple mean-structure models. For each sample, we specified two models that differed only in their mean structure. Both models included a saturated covariance structure, with all variances and covariances among the observed variables estimated freely. In the more parsimonious model, the means of each construct were constrained to be equal across waves. In the comparison model, the means were estimated freely at each wave. Model fit was compared using $RMSEA_D$ (Savalei et al., 2024). An $RMSEA_D$ above the cutoff of .08 was taken to indicate that the equal-means model should not be retained and that temporal mean trends should be modeled in the longitudinal analyses.

Visual inspection suggested systematic downward trends for both constructs in all three samples (Figure S1). RMSEAD exceeded the cutoff in all three samples (Bern: $RMSEA_D$ = .386; Zurich: $RMSEA_D$ = .370; Trier: $RMSEA_D$ = .200), indicating that the equal-means model should not be retained.

Part 4: Comparison of Nondetrended and Detrended RI-CLPMs

Following Falkenström et al. (2026), we compared, for each sample, a nondetrended RI-CLPM, in which the intercepts of the observed variables were constrained to be equal across waves, with a detrended RI-CLPM, in which these intercepts were estimated freely at each wave. The nondetrended model showed poor fit in all three samples, and its unconstrained version failed to converge in the Bern sample (Table S1). $RMSEA_D$ far exceeded the cutoff of

[1] The stored pretreatment item values correspond to the transformation $y = 1 + 0.75x$, where x denotes the original BSI response coded from 0 to 4.

.08 in all three samples ($RMSEA_D$ = .210–.421; Table S2), indicating that the nondetrended specification led to a meaningful loss of fit.

Part 5: Exploratory Analyses: Latent Curve Models With Structured Residuals

Convergence

We refrained from interpreting the estimates from both the Bern and Zurich samples because the constrained and the unconstrained LCM-SRs either failed to converge properly or yielded inadmissible parameter estimates. In the Bern sample, the unconstrained model did not converge, and the constrained model yielded inadmissible parameter estimates: all four growth-factor variances were negative, and three latent correlations were out of range. In the Zurich sample, both models converged, but their estimates were inadmissible: the correlation between the random slopes was out of range in the constrained (r = 1.40) and the unconstrained model (r = 1.25), and the unconstrained model additionally yielded a negative residual variance. Estimates from these two samples are therefore not interpretable.

In the Trier sample, both LCM-SR specifications converged properly. The constrained model was retained because $RMSEA_D$ was below the cutoff of .08 ($RMSEA_D$ = .017), indicating that the equality constraints on the autoregressive and cross-lagged effects were acceptable. However, the constrained model yielded a near-boundary solution. The growth-factor covariance matrix was marginally non-positive definite (minimum eigenvalue = −.0003), and the correlation between the random slopes was high (r = .981). No latent variances were negative, all pairwise latent correlations were within range, and no additional numerical estimation problems were identified. We therefore interpreted the estimates from this model cautiously.

Estimates in the Trier Sample

In the constrained model, the autoregressive effects were significant for both incongruence (β = .18) and distress (β = .10; Table S4). Incongruence had a significant prospective effect on distress (β = .06), whereas the reverse effect from distress to incongruence was not significant

($\beta = .01$). In the unconstrained model, four of the five effects of incongruence on subsequent distress were positive (the fifth was near zero), although only one was significant. None of the five reverse effects were significant.

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Table S1

Model Fit for the Unconstrained and Constrained Nondetrended Random-Intercept Cross-Lagged Panel Models of Incongruence and Distress

	Unconstrained				Constrained				
Sample	χ^2	*df*	CFI	RMSEA [90% CI]	χ^2	*df*	CFI	RMSEA [90% CI]	$RMSEA_D$
Bern trial	—	—	—	—	137.3	23	.67	.222 [.187, .258]	—
Zurich trial	135.9	15	.76	.236 [.200, .273]	148.7	23	.75	.194 [.165, .224]	.064
Trier	752.8	47	.94	.103 [.097, .110]	811.4	63	.94	.092 [.086, .097]	.043

Note. CFI = comparative fit index; RMSEA = root-mean-square error of approximation; CI = confidence interval; $RMSEA_D$ = RMSEA associated with the difference test. The nondetrended RI-CLPM constrains intercepts of observed variables to be equal across waves (see Method). In the constrained models, the autoregressive and cross-lagged effects were constrained to be equal across measurement intervals, whereas in the unconstrained models, these effects were estimated separately for each interval. The fit of the constrained and unconstrained models was compared using the $RMSEA_D$ index (Savalei et al., 2024). When $RMSEA_D$ is smaller than .08, the results indicate that the more parsimonious model (i.e., the constrained model) should be preferred. The em dash indicates that the model did not converge or did not converge properly (see Results section for explanation).

Table S2

Model Fit for the Detrended and Nondetrended Random-Intercept Cross-Lagged Panel Models of Incongruence and Distress

	Detrended				Nondetrended				
Sample	χ^2	*df*	CFI	RMSEA [90% CI]	χ^2	*df*	CFI	RMSEA [90% CI]	$RMSEA_D$
Bern trial	24.1	17	.98	.064 [.000, .119]	137.3	23	.67	.222 [.187, .258]	.421
Zurich trial	10.2	17	1.00	.000 [.000, .034]	148.7	23	.75	.194 [.165, .224]	.390
Trier	180.2	53	.99	.041 [.035, .048]	811.4	63	.94	.092 [.086, .097]	.210

Note. CFI = comparative fit index; RMSEA = root-mean-square error of approximation; CI = confidence interval; $RMSEA_D$ = RMSEA associated with the difference test. The nondetrended RI-CLPM constrains intercepts of observed variables to be equal across waves (see Method). In both types of models (i.e., the detrended and nondetrended) the autoregressive and cross-lagged effects were constrained to be equal across measurement intervals. The fit of the detrended and nondetrended models was compared using the $RMSEA_D$ index (Savalei et al., 2024). When $RMSEA_D$ is smaller than .08, the results indicate that the more parsimonious model (i.e., the nondetrended model) should be preferred.

Table S3

Model Fit for the Unconstrained and Constrained Latent Curve Model With Structured Residuals of Incongruence and Distress

	Unconstrained				Constrained				
Sample	χ^2	*df*	CFI	RMSEA [90% CI]	χ^2	*df*	CFI	RMSEA [90% CI]	$RMSEA_D$
Bern trial	—	—	—	—	29.9	14	.95	.106 [.052, .159]	—
Zurich trial	12.5	6	.99	.086 [.000, .154]	15.4	14	1.00	.027 [.000, .087]	.000
Trier	201.9	38	.99	.055 [.048, .063]	224.5	54	.99	.047 [.041, .054]	.017

Note. CFI = comparative fit index; RMSEA = root-mean-square error of approximation; CI = confidence interval; $RMSEA_D$ = RMSEA associated with the difference test. The fit of the constrained and unconstrained models was compared using the $RMSEA_D$ index (Savalei et al., 2024). When $RMSEA_D$ is smaller than .08, the results indicate that the more parsimonious model (i.e., the constrained model) should be preferred. The Bern and Zurich LCM-SR models were not further interpreted because they either did not converge properly or yielded inadmissible parameter estimates (including negative growth-factor variances and out-of-range correlations in the constrained Bern model and an out-of-range correlation between the random slopes in both Zurich models). Em dash indicates that the model did not converge properly.

Table S4

Cross-Lagged and Autoregressive Effects Between Incongruence and General Psychological Distress in Latent Curve Models With Structured Residuals

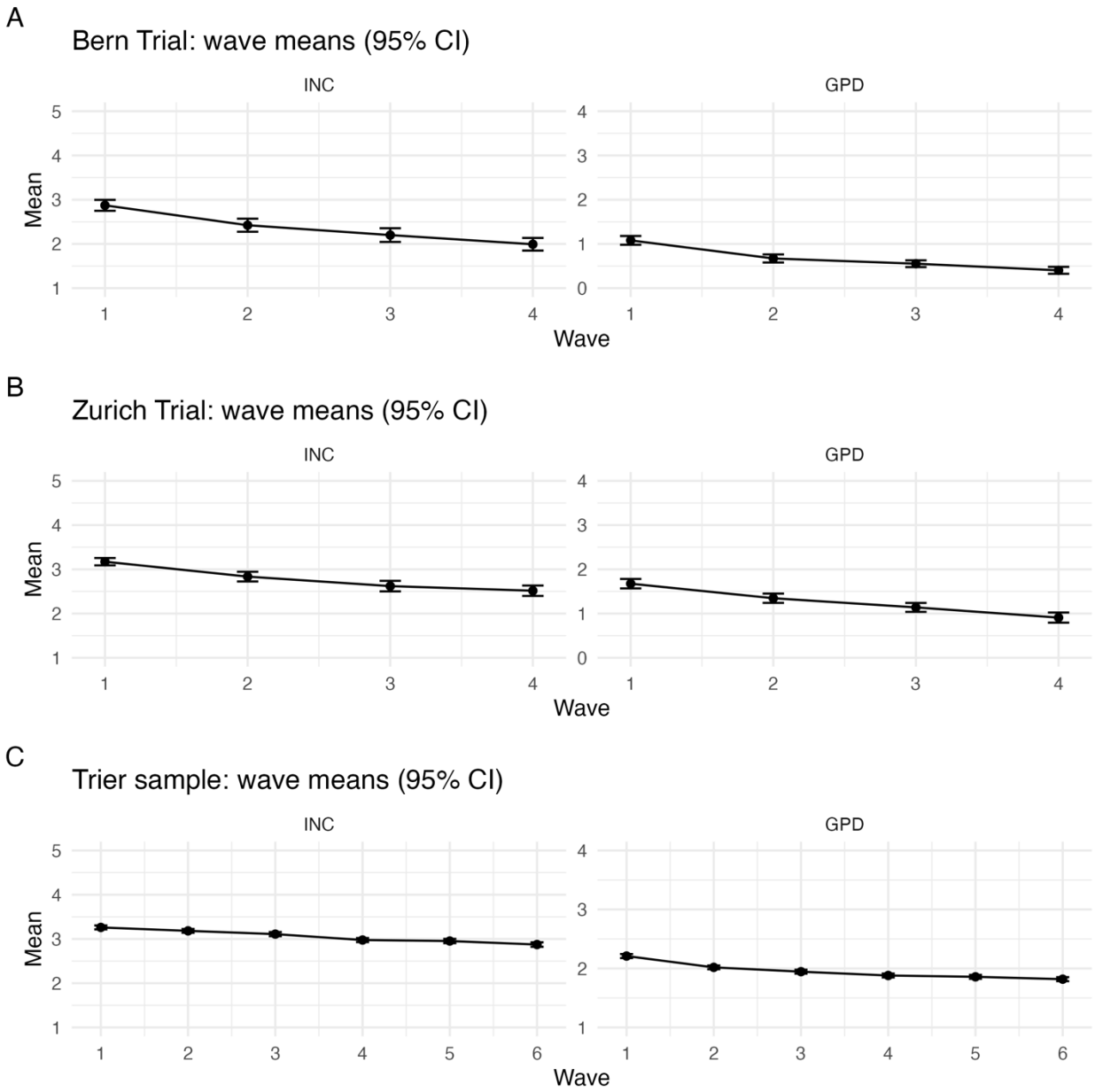
		Cross-lagged effects		Autoregressive effects	
Sample	$r_{INC,GPD}$	INC → GPD	GPD → INC	INC → INC	GPD → GPD
Bern trial	—	—	—	—	—
Zurich trial	—	—	—	—	—
Trier	.83* [.79, .87]	.06* [.01, .10]	.01 [–.03, .06]	.18* [.12, .24]	.10* [.05, .15]

Note. The table shows standardized coefficients. INC = motivational incongruence; GPD = general psychological distress; $r_{INC,GPD}$ = correlation between the random intercepts of incongruence and psychological distress. The cross-lagged and autoregressive effects were constrained to be equal across measurement intervals. Note that we only report estimates from models that converged and that we refrained from interpreting the estimates from both the Bern and Zurich samples (see Supplemental Material Part 5). Em dash indicates that estimates are not reported because no admissible solution was obtained. Values in brackets are 95% confidence intervals. The autoregressive and cross-lagged coefficients and their confidence intervals are averaged across waves.

* $p < .05$

Figure S1

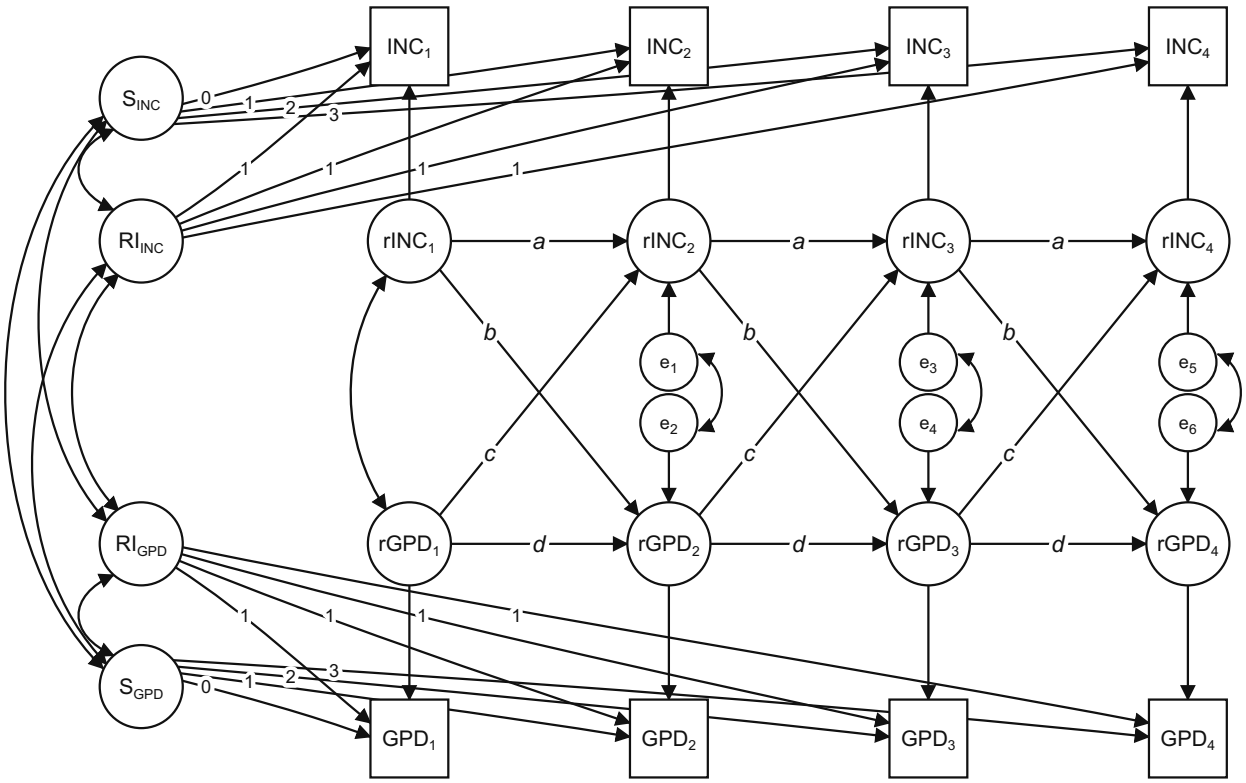
Means of Motivational Incongruence (INC) and General Psychological Distress (GPD)



Note. Points represent wave means estimated using full-information maximum likelihood, and error bars represent 95% confidence intervals. INC = motivational incongruence; GPD = general psychological distress. INC scores ranged from 1 to 5 in all three samples. GPD scores ranged from 0 to 4 in the Bern and Zurich samples and from 1 to 4 in the Trier sample.

Figure S2

Latent Curve Model with Structured Residuals



Note. The figure illustrates the latent curve model with structured residuals used to examine prospective effects between motivational incongruence (INC) and general psychological distress (GPD), shown for four waves as in the Bern and Zurich trial samples. The model for the Trier sample followed the same structure but included six waves. Residual variances are denoted as e1 to e6. Letters a, b, c, and d represent equality constraints on paths labeled identically. The residualized scores (r) of the constructs are time-varying components. The random intercepts (RI) are time-invariant latent factors capturing stable between-person differences in each construct. The latent linear slope factors (S) capture individual differences in each construct's rate of change over time.